

Mobile Programming



Node.js Question Bank

Section A: Theoretical/Conceptual Questions

1. What is Node architecture and what does it include?
2. How does the event loop manage requests in Node.js?
3. What is Callback Hell and what are some solutions to it?
4. What is the difference between a promise and async/await in Node.js?
5. How can you add custom middleware in Node.js for handling requests?
6. What is middleware chaining and how does it work in Node.js?
7. How can you write the words "one Two Three" in the console when given the variable a = oneTwoThree in Node.js?
8. How to find the second maximum value from an array in JavaScript? Can you provide a solution with and without using built-in functions?
9. How can you merge three objects in JavaScript? Can you provide a solution with and without using built-in functions?
10. How to merge multiple objects into a single object and remove duplicates in JavaScript? Can you provide a solution with and without using built-in functions?
11. Can you explain what the Event Loop is in Node.js and how it works?
12. Can you explain what Promises are in JavaScript and provide an example of making nested Promises?
13. How can you avoid using nested Promises in JavaScript?
14. Can you explain what async/await is in JavaScript and how they are used for handling asynchronous operations?
15. Explain express middlewares?
16. How will you implement the middleware chain?
17. What is the difference in var and let?
18. How would you clone an object?
19. How would you handle doing large processing in node.js? For example sending a newsletter to millions of people?
20. Explain the child process?
21. What do you know about Promise.all?
22. What are the different strategies to merge two arrays?
23. Which database did you use with node.js?
24. Explain when you would need to implement caching?
25. What is an XSS (Cross-site scripting) attack and how does it work?

26. What is SQL injection and how can it be prevented?
27. What is the difference between blocking and non-blocking code?
28. What are middlewares in Node.js?
29. What is an error-first callback in Node.js?
30. What are the alternative ways to handle asynchronous code in Node.js?
31. If we have multiple promises, how can we ensure that they are all settled before further processing?
32. What are the different security parameters to keep in mind when writing a REST API?
33. What is a Cross-Origin Request and how can it be handled?
34. What is an index in a database and how does it work?
35. What is aggregation in MongoDB?
36. What is an embedded document in MongoDB?
37. What is the data type of `_id` in MongoDB?
38. What is Projection in MongoDB?
39. How would you perform partial text search in MongoDB?
40. What are stored procedures in MongoDB?
41. What is the spread operator in JavaScript?
42. How would you find the second largest number in an array in JavaScript?
43. Explain the differences between callback, promise, and generator functions in JavaScript.
44. How would you find the number of occurrences of a character in a string in JavaScript?
45. Can you introduce yourself and talk about your past experience and the technologies you have worked on?
46. What is queueing in Node.js and how does it work?
47. Can you explain the concept of the event loop in Node.js?
48. How do you scale an application in Node.js?
49. What are cron jobs and how do they work in Node.js?
50. Can you explain the concept of promises in JavaScript, and provide a code snippet example?
51. Can you explain the concept of `async/await` in JavaScript, and provide a code snippet example?
52. How to implement inheritance in JavaScript?
53. What are Promises in JavaScript? Why use them?
54. Can you provide a code example for an `async` request handled with promises?
55. What is the difference between callbacks and promises?
56. Can you explain how to print the outermost elements of a 4x4 matrix in a spiral order?
57. What is callback hell? Can you give an example?
58. How do you remove duplicate elements from an array in JavaScript? Provide an example.
59. What is the syntax to add a new element at the beginning of an array in JavaScript?
60. How do you merge two arrays in JavaScript? Provide an example.
61. What is variable hoisting in JavaScript? How does it work? Explain the concept of scopes in JavaScript.
62. What is the purpose of `process.nextTick` in Node.js and how does it work? Provide an example code snippet to demonstrate its usage.
63. Can you share a recent issue you faced in your project and explain how you went about solving it?
64. What is the purpose of the `cluster` module in Node.js? How does it work?
65. What is a child process in Node.js? How is it different from a cluster?
66. Create a Hapi Server and write an endpoint that returns current server time.
67. What is Promise in Javascript? How to execute promises in parallel?

68. What is the difference between `async-await` and `Promise` in Javascript?
69. Write a program in Javascript to find the second maximum element in an array.
70. Write a program in Javascript to merge two sorted arrays into a sorted form.
71. What is the event loop in Node.js?
72. Write a program in Javascript to shift the *k*th element of an array from the start to the end.
73. Create two Node.js services that communicate to exchange generic objects. Each service should have three API endpoints, for a total of six. Use a database to save the data.
74. What is the threading model of Node.js? Is Node.js single threaded or does it support multi-threading? Can you explain the event-driven architecture of Node.js and how it helps in handling multiple requests efficiently?

Section B: Coding Questions

75. How can you count the number of occurrences of a particular character in a JavaScript String, and what are some different approaches that can be used to achieve this?
76. What is a possible approach to find the most frequently occurring character in a JavaScript String, and how can you implement this solution in JavaScript programming?
77. What is an approach to add the individual digits of a given integer in JavaScript programming, and how can you implement this solution to calculate the sum of digits for an integer such as 567 and return the result of 9?
78. Write a JavaScript program to find the largest subarray in an integer array that has a sum of 0. Can you explain your approach to solving this problem and provide a step-by-step explanation of your code? How does your program handle edge cases, such as when there is no subarray with a sum of 0 or when the input array is empty?

Eg: Input = { 3, 4, -7, 3, 1, 3, 1, -4, -2, -2 }

```
{ 3, 4, -7 }  
{ 4, -7, 3 }  
{ -7, 3, 1, 3 }  
{ 3, 1, -4 }  
{ 3, 1, 3, 1, -4, -2, -2 }  
{ 3, 4, -7, 3, 1, 3, 1, -4, -2, -2 } // This should be the output
```

79. Can you provide a program in JavaScript that finds all the subarrays in an array whose sum is zero? How does your implementation work? Can you explain the time and space complexity of your solution? Are there any optimizations that can be made to improve the efficiency of your program?
80. Write a JavaScript program to find the sum of all the numbers in a given string. For example, if the input string is "1ab2d4hj6", the program should output the sum of numbers in the string, which is 13.

The program should take a string as input from the user and use regular expressions to extract all the numbers from the string. It should then iterate over the extracted numbers and add them up to get the final sum. If there are no numbers in the string, the program should output 0.

81. Write a JavaScript program that takes two strings as input and finds the number of occurrences of the second string in the first string.

Example:

Input:

String 1: "hello world"

String 2: "l"

Output:

Number of occurrences of "l" in "hello world" is 3

82. Write a JavaScript program to find the sum of all the numbers in a given string. Your program should take a string as input from the user and then find all the numbers in the string. It should then add up all the numbers and print the sum to the console.

Here are some requirements for your program:

The string should be read from the console using the Scanner class.

Your program should use regular expressions to find all the numbers in the string.

The sum of the numbers should be calculated using a loop.

If there are no numbers in the string, your program should print a message saying so.

Example:

Input: "Hello 123 world 456"

Output: 579

Write test cases on the same program by using the assertion method in JUnit.

83. Write a JavaScript program to find the sum of all the numerical digits in a given string s1. The string can contain alphabets and digits in any order, but the digits are guaranteed to be consecutive. Your program should output the total sum of all the digits in the string.

For example, given the string s1 = "abc4ui78fg95", the program should output sum = 177, since the digits in the string are 4, 7, 8, 9, and 5, and their sum is $4 + 7 + 8 + 9 + 5 = 33$.

84. Given a matrix of integers, write a program to modify the matrix such that if any element of the matrix is 0, then the entire row and column of that element should be set to 0. Implement this program using JavaScript.

The program should take the following input:

The matrix, represented as a two-dimensional array of integers.

The program should perform the following tasks:

Check for the presence of 0 in the matrix and record the positions of the 0 elements.

Modify the matrix by setting the entire row and column to 0 if any element in the row or column contains a 0.

The program should return the modified matrix as output.

For example, given the matrix:

{ 1, 2, 3 }

{ 4, 0, 6 }

{ 7, 8, 9 }

The program should modify the matrix to:

```
{ 1, 0, 3 }  
{ 0, 0, 0 }  
{ 7, 0, 9 }
```

And return the modified matrix as output.

85. Write a JavaScript program to find all the duplicate elements in an integer array with less time complexity.

Example input:

```
int[] arr = {4, 2, 4, 5, 2, 3, 1, 1, 6, 7, 7};
```

Expected output:

Duplicate elements in the given array are:
4 2 1 7

Constraints:

The input array can have a maximum of 10^5 elements.

The program should have a time complexity of $O(n)$, where n is the size of the input array.

86. Write a JavaScript program to reverse an array without using any inbuilt methods.
87. Can you explain the functionality of the following code?

```
function camelCase(a){  
  var b = a.split(/(?=[A-Z])/i);  
  
  for(i = 0; i < b.length; i++) {  
    console.log(b[i]);  
  }  
}
```

What does this function do? Can you explain the role of each line of code?

88. Write a SQL query to fetch department details for a given employee ID?
89. Write a program in JavaScript to find if there exist two numbers in an array whose sum is equal to a given target sum. The program should take an array of integers and the target sum as input and return a boolean value indicating if there exist two numbers in the array whose sum is equal to the target sum.
90. Write a JavaScript program to display the highest prime number.
91. Program to find the largest number in a given input array and replace it with a given number/string.
In an array take alternative numbers and reverse them. Put it again in the place of array again
e.g., arr = [1,2,3,4,5,6,7,8]

`newarr = [1,8,3,6,5,4,7,2]`

92. Given an unsorted array `a = [3,1,5,2,7]`, find alternate elements "3,5,7", reverse them "7,5,3" and place them in the original array

Expected result `[7,1,5,2,3]`

93. Find second largest number `[3,1,5,2,7]`
94. Can you write a program in JavaScript to find the largest number in a given input array and replace it with a given number or string?
95. Out of given 3 arrays with n number of elements, find out the common element(s) for all 3 arrays.
96. From a given array, make n elements rotate.

eg. given Array:`[21,54,11,35,4,18]`, index:3; output Array: `[4,18,21,54,11,35]`

97. From an array, find out the elements, and their repetition and print as element=count.

eg.

Input Array: `[2,1,3,4,6,5,4,7,5,6,8,2,3,6,2,6,4,4,7,3,4,5,6]`

Output:

`2=3`

`1=1`

`3=3`

`4=5`

`6=5`

`5=5`

`7=2`

`8=1`

98. Print the right diagonal of the matrix

Input:

1 2 3 4

5 6 7 8

9 10 11 12

13 14 15 16

Output:

4 7 10 13

99. String= "abaade", search for a character "a" in the given string and return its occurrence.
100. How will you remove duplicate elements from an array?
101. Can you provide a sample program in JavaScript that takes an array as input and generates a new array containing only unique elements, and explain how it works in detail?
102. How can you extract all the keys from a nested object in JavaScript, given an object 'object = {id:1 , address : {primaryAddress:'abc'} }', and return an array of keys as ['id','address','primaryAddress']? Also, how would you remove duplicates from an array '[1,1,2,2,3,4,4,5]' in JavaScript?
103. Write a JavaScript function that takes an array of numbers as input and returns a new array where each element at index i is the sum of all the elements in the input array up to index i. For example, if the input array is [1, 2, 3, 4], the output should be [1, 3, 6, 10].
104. Given an array of strings, write a JavaScript function that returns the longest string from the array. For example, if the input array is ["i am in pune", "i am in hyderabad", "location"], the output should be "i am in hyderabad".
105. Write a JavaScript function to find the total cost and discount given an object that contains nested objects representing items and their attributes.

Input:

The input object will have a nested structure, with each item represented as a key-value pair. The attributes of each item are represented as key-value pairs within the object.

Output:

The output should be an object with two properties, cost and disc, representing the total cost and discount, respectively, of all items in the input object.

Example:

```
const obj = {
  id1: {
    name: "adf",
    cost: 10,
    disc: 20,
  },
  id2: {
    name: "lkfd",
    cost: 20,
    disc: 10,
    items: {
      id3: {
```

```

        name: "sfd",
        cost: 10,
        disc: 20,
    },
},
},
};

const result = getTotalCostAndDiscount(obj);

console.log(result); // { cost: 40, disc: 50 }

```

Function Signature:

```

function getTotalCostAndDiscount(obj) {
    // function body
}

```

106. You are given a sorted array of integers and a target difference. Write a function `findPair` that takes in the array and the target difference and returns an array of the pair of numbers that give the target difference. If no such pair is found, return an empty array.
107. Write a program that finds the longest string in an array and returns the number of words in that string. The input array will be of the form:

```

const sentences = ["alice and bob love leetcode", "i think so too", "this is great thanks very much"];

```

The program should output the longest sentence along with the number of words in that sentence.

Additionally, write a program that takes an input array of numbers and returns an output array where the first element of the output array is the first element from the input array, the second element of the output array is the sum of the first and second elements from the input array, the third element of the output array is the sum of the previous element ($1 + 2 = 3$) and the next element ($3 + 3 = 6$), and the fourth element is the sum of the previous element ($3 + 3 = 6$) and the next element ($6 + 4 = 10$). The input array will be of the form:

```

const inputArray = [1, 2, 3, 4];

```

The program should output an array of the form:

```

outputArray = [1, 3, 6, 10];

```

Section C: Soft Skills Questions

1. Tell me about a project you worked on where you had to communicate technical information to a non-technical audience.
2. How do you prioritize and manage your workload?
3. Can you describe a time when you had to work with a difficult team member?
4. How do you handle criticism of your code or work?
5. Can you describe a time when you had to work under pressure to meet a deadline?
6. How do you ensure that your code is maintainable and scalable?
7. Tell me about a time when you had to learn a new technology or programming language quickly.
8. How do you keep up-to-date with the latest developments in software development?
9. Can you describe a time when you had to work on a project that required collaboration with multiple teams?
10. How do you approach problem-solving in your work?
11. Can you give an example of how you have applied creativity in your work as a software developer?
12. Can you describe a time when you had to work on a project with a tight budget?
13. How do you ensure that your code is properly documented?
14. Tell me about a time when you had to work with a client or end-user to understand their needs and requirements.
15. How do you approach debugging complex software issues?
16. Can you describe a time when you had to make a trade-off between code quality and speed of delivery?
17. How do you collaborate with other developers on a project?
18. Tell me about a time when you had to mentor or train other developers.
19. Can you describe a time when you had to persuade a team to adopt a new technology or approach?
20. How do you approach testing and ensuring the quality of your code?